



Pedestrian road-crossing willingness as a function of vehicle automation, external appearance, and driving behaviour

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ABSTRACT

This paper discusses whether the knowledge of the driving mode of an approaching vehicle (manual vs. automated) influences pedestrians' decisions while crossing a street. Additionally, the paper explores how different appearances and driving behaviours of vehicles interact with driving mode in affecting pedestrians' road-crossing behaviours. In a video-based experiment with sixty participants, two vehicles with different appearances (a BMW 3 and a Renault Twizy) were presented as either manually-driven or automated vehicles. Both vehicles displayed either yielding or non-yielding behaviour on a straight road devoid of other traffic. Participants were asked to indicate whether they would cross the street in front of the approaching vehicle, at different distances ranging from 45 m to 1.5 m. The results showed that there was no significant influence of the knowledge of the driving mode (manually-driven vs automated) on pedestrians' willingness to cross the street at any distance. The vehicle's behaviour (whether it is maintaining speed or yielding) played a dominant role in pedestrians' decision to cross a road, and this was similar for both modes and both vehicles, at all distances. However, results suggested that in situations and at distances when the intent of the vehicle was not fully clear by the behaviour of the car alone, there were differences between the two vehicles at certain distances, which could be attributed to the differences in their appearance such as size, aggressiveness and novelty. A futuristic-looking vehicle inspired less confidence in road-crossing situations compared to an ordinary-looking vehicle. Additionally, a novel and futuristic-looking vehicle appeared to make it easier for people to believe that it is an automated vehicle. We conclude by discussing design implications for the development of external HMIs automated vehicles.

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1. Introduction

Rapid changes in automotive technology are resulting in drastic changes in the driving task, by which the decisions and control over what the vehicle does are increasingly shifted from the driver to the system. Much attention is already being devoted to what this means for the driver, in terms of trust, transparency, and how the interface can be adjusted to supporting non-driving-related tasks (Kun, 2018). Recently, research and development in academia and industry has also started to give attention to the question of what vehicle automation means for the interaction with other road users (Lundgren et al., 2016; Merat, Louw, Madigan, Wilbrink, & Schieben, 2018; Mirnig, Perterer, Stollnberger, & Tscheligi, 2017). There have been

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numerous studies and reports which highlight the general lack of trust and acceptance of automated vehicles among people. [Bazilinskyy, Kyriakidis, and De Winter \(2015\)](#) found from their crowdsourcing study that opinions on trust concerning automated vehicles were vastly divided, while other reports showed that people are generally not receptive to the idea of automated driving, and are cautious and hesitant to the idea of sharing road spaces with driverless vehicles ([Hutson, 2017](#); [Nordhoff, De Winter, Kyriakidis, Van Arem, & Happee, 2018](#); [Smith & Anderson, 2017](#)). Recently, [Hulse, Xie, and Galea \(2018\)](#) found that automated cars were perceived as a “somewhat low risk” form of transport. Another interview study of pedestrians who had interacted with Uber automated vehicles found that an ‘innocent until proven guilty’ mentality is prevalent with respect to automated vehicles and trust ([Reig, Norman, Morales, Das, Steinfeld, & Forlizzi, 2018](#)). However, the authors also note that there was a sizeable number of participants yet to be convinced: the belief that automated driving technology is not yet mature, or a general lack of knowledge about what automated driving technology is, lead to mistrust in several participants. People also associated automated vehicles with artificial intelligence (AI), and both are perceived by some to be detrimental to society. This was particularly observed in the incidents involving vandalism and harassment of Waymo automated vehicles in Arizona amidst the general mistrust towards automated vehicles ([Kautonen, 2019](#); [Romero, 2018](#)). Given these split opinions, an interesting question is how pedestrians will behave in the interaction with automated vehicles as compared to manually driven vehicles in the context of a specific task such as crossing a street in view of an approaching vehicle.

When pedestrians want to cross the street while a vehicle is approaching, they need to decide whether to cross the street in front of the approaching vehicle or wait until it has passed. When the vehicle in question is being driven in an automated manner, the question arises whether the perceptions that factor into a pedestrian's decision-making differ from when interacting with a manually-driven vehicle. Given the novelty of automated driving and automated vehicles not being mainstream, we anticipate a certain hesitation around automated vehicles born out of lack of trust or familiarity.

A few studies conducted in this direction show some preliminary results. [Rothenbücher, Li, Sirkin, Mok, and Ju \(2016\)](#) tested their “ghost driver” platform by hiding a human driver inside a seat suit in a car labelled as an automated vehicle. At pedestrian crossings and roundabouts, they found that the Wizard-of-Oz automated vehicle did not alter the interactions and road-crossing behaviour of pedestrians as long as the vehicle did not behave in an unpredictable manner. Furthermore, a study conducted by [Rodríguez Palmeiro et al. \(2018\)](#) reported similar results: when pedestrians interacted with Wizard-of-Oz automated vehicles where drivers were distracted by other activities or when a car was clearly marked as self-driving, pedestrians' willingness to cross did not change. However, since the manipulations in these studies were done by addition of stickers to passenger cars that were otherwise normal in appearance, the question arises whether the participants read the stickers or achieved enough immersion in the studies.

Furthermore, apart from pedestrian-related factors such as mobility, assertiveness, culture, and possibly the context (being in a hurry or not, as well as location on the road), vehicle-related factors, both behavioural (distance, speed and acceleration) and appearance (size, colour, and styling as influenced by outer looks that hint at the vehicle's sociability and power) play a role in pedestrians' crossing decisions ([Klatt, Chesham, & Lobmaier, 2016](#); [Windhager et al., 2008, 2012](#)). Specifically, the appearance of a vehicle can lend itself to stereotypes about the kind of people who own them, and their driving behaviour ([Davies & Patel, 2005](#); [Davies, 2009](#)). Studies have shown that in the case of manually-driven vehicles, this, in turn, may affect pedestrians' crossing behaviours in front of cars ([Klatt et al., 2016](#)). In case of manually-driven vehicles, the appearance of a vehicle can also lend itself to stereotypes about the kind of people who own them, and their driving behaviour ([Davies & Patel, 2005](#); [Davies, 2009](#)). This, in turn, affects pedestrians' crossing behaviours in front of cars ([Klatt et al., 2016](#)). A pedestrian may look at a vehicle and instantly make assumptions about the vehicle's behaviour based on a complex combination of vehicle and driver characteristics, in terms of its assertiveness, potential to yield, etc., and consequently determine how to interact with it. Contrarily, given that an automated vehicle is driven by computer algorithms and not a human driver, the question arises whether vehicle appearance at all matters in the context of pedestrian interactions with it, since stereotypes may not apply to a robotic system as it does to humans. It is thus also interesting to investigate how significantly different appearances and driving behaviours of vehicles interact with the driving mode of the vehicle.

From these considerations, we derive the following main hypotheses:

H1. The perceived driving mode of the vehicle (manual vs automated) affects the willingness of pedestrians to cross the street in favour of a manually-driven car – pedestrians may be more hesitant to cross around an automated vehicle at the same distance due to a novelty effect, lack of trust, or other factors.

H2. The driving behaviour of the vehicle affects the willingness of pedestrians to cross the street in favour of a vehicle demonstrating yielding behaviour, all other factors remaining constant.

H3. The effect of the appearance of a vehicle will depend on its automation status:

H3a. The appearance of a manually-driven vehicle will affect the road-crossing willingness of pedestrians in favour of a smaller, friendlier appearance, all other factors remaining constant.

H3b. The appearance of automated vehicles will not have a significant effect on the road-crossing willingness of pedestrians, all other factors remaining constant.

The hypotheses were evaluated in a study deploying video clips of approaching vehicles. While a video-based study takes away the dynamics of the interaction between the pedestrian and the vehicle, it ensures for solid experimental control. Furthermore, it has been shown that time arrival estimates do not differ between video clips and real-life situations ([Recarte,](#)

Conchillo, & Nunes, 2005), which justified that a video-based study in this context preserves the ecological validity of the insights.

2. Method

2.1. Selection of vehicles

The scope of the experiment was limited to the interaction between vehicles and pedestrians (and not cyclists or non-automated, human-driven vehicles), in which we measured a pedestrian's willingness to cross the road in the presence of the approaching vehicle, whether automated or manually driven. Besides quantitative data, the experiment also aimed to get qualitative insights on people's reaction and behaviour to an automated vehicle in contrast to a manually-driven vehicle. The main manipulation was through instructions to participants: they were told that the approaching vehicle was either an automated vehicle or a manually-driven vehicle.

The project which this study is a part of focuses on creating an automated vehicle and using it to study different aspects of vehicle automation, including technological and human factors issues. The carrier for the project is a Renault Twizy. A Renault Twizy has a rather extraordinary appearance, and, by virtue of its appearance, may already interfere with the mental models of pedestrians and cause some uncertainty in road-crossing situations. Therefore, a second car had to be chosen that opposed the appearance-related characteristics of the Twizy for inclusion in the experiment, so that the results of diametrically opposite dimensions within vehicles' appearances can be compared, and their interactions with vehicle driving behaviour and driving mode studied.

In order to assess how the Renault Twizy was perceived, and how it compared to other vehicles regarding its "sociability" and "power" as explained in the work of Windhager et al. (2008, 2012), a few other commonly available vehicles were chosen that varied across these dimensions. Thus, an inventory of 8 different vehicles was created based on parameters such as corners of windshield, extremes of headlight position, dimensions of grill (Windhager et al., 2008) regarding cars that we believed represented different degrees of sociability and power, as well as novelty of appearance (BMW 3 series F30, Google self-driving car, Nissan 370z, Renault Twizy, Toyota Prius XW30, Volkswagen Beetle A5, Volkswagen Golf MK6, and Volkswagen Golf MK7). We sought confirmation of our impressions of the perceptions of the sociability and power, and additionally, the novelty and futuristic-ness of the vehicles' appearance through a survey. Images of the front view of these 8 vehicles juxtaposed with a picture of a pedestrian for scale of size were shown in a randomized order in the survey, and the respondents were asked to rate the vehicles on various external-appearance-related dimensions using scales (Table 1). To explore the perception of each of these dimensions, six scales were formulated with semantic anchors specifically chosen to represent two extremes of the three appearance-related dimensions (sociability, power, and novelty). This was done by framing questions in a manner that oriented the participants towards the scenario of a pedestrian facing the vehicle. The responses revealed general perception towards the different vehicles in the context of a pedestrian's road-crossing situation in the presence of the corresponding approaching vehicle.

In the cars shown, any visible logos on the car that identified the vehicle as a particular make or model were purposefully and carefully hidden. This was done to achieve a general sense of anonymity about the vehicles and prevent bias based on brand prejudice. While this would not have deterred well-seasoned car enthusiasts from recognizing the car from its general shape and appearance, this was a general precaution applied within the practical limitations of the experiment to mitigate any confounding influence. Furthermore, according to Davies and Patel (2005), white as a colour in cars score relatively neutral in perception of aggressiveness of a vehicle. Since the objective was to elicit judgements about the cars strictly by the design/appearance of the vehicle, variation of colour was removed to eliminate a possible confounding factor. Hence, all the cars shown were white in colour (except for the Renault Twizy, which had white body panels and black bumpers) to prevent further implicit prejudice based on car colour. The survey was distributed over professional and social networks and allowed to run until at least 100 responses were received (final sample size $N = 102$). We used the survey results to calibrate the appearance of the different cars in order to have an objective measure of how friendly and futuristic the vehicles looked. The responses were collected on a 10-point scale based on perceptions of various attributes of the vehicles. Among the vehicles evaluated in the survey, the vehicle that was rated as the most "different" in appearance-related dimensions from the Twizy was selected, which was a BMW 3 series. Thus, the vehicles finally used for the study were:

Table 1

Mean and standard deviation of survey responses for various attributes of the selected vehicles, $N = 102$.

	Scales	BMW 3 series (mean, sd)	Renault Twizy (mean, sd)	t(101), p
1	Cute – Aggressive (1–10)	7.31, 1.9	3.29, 2.05	13.989, <0.000
2	Submissive – Dominant (1–10)	7.85, 1.64	3.29, 1.94	17.501, <0.000
3	Friendly – Angry (1–10)	7.24, 1.89	3.72, 2.04	12.378, <0.000
4	Yielding – Assertive (1–10)	7.58, 1.62	3.78, 2.12	13.878, <0.000
5	Ordinary – Futuristic (1–10)	4.71, 1.9	7.8, 1.99	10.937, <0.000
6	Commonplace – Alternative (1–10)	4.27, 1.89	7.98, 2.11	12.744, <0.000

- Vehicle V1 – BMW 3 series: Representative of a commonplace and ordinary, but aggressive and angry-faced vehicle.
- Vehicle V2 – Renault Twizy: Representative of a futuristic and alternative, but cute and friendly-faced vehicle.

The mean and standard deviation of the responses in the survey for the selected cars are summarized in [Table 1](#).

2.2. Video-based experiment

2.2.1. Task

At its simplest, the core task in this experiment was for a participant assuming the role of a pedestrian to watch a number of randomized video segments of an approaching vehicle on an unmarked, otherwise-empty road when the vehicle was at predetermined distances away from them, and indicate their willingness to cross the road at specific moments.

2.2.2. Materials

Recognizing the difficulty in controlling the parameters of the moving car and the timing of the pedestrian's decision, the experiment was carried out as a video-based study instead of in the field. Videos were captured where the two vehicles exhibited two different behaviours, and showed two different driving modes, as explained below.

- Behaviour B1: Steady driving at a speed of 50 km/h
This behaviour is an indication that the car is going at a constant speed and does not intend to slow down or yield. This is representative of a non-yielding behaviour.
- Behaviour B2: Deceleration from 50 km/h to a full stop in front of the pedestrian
This behaviour indicates that the car has spotted the pedestrian and is intentionally stopping. The braking was gentle and purposeful (deceleration rate of 1.8 m/s^2 , braking started 54 m away from the pedestrian). This is representative of yielding behaviour.

To enhance the effect of the primary manipulation, by which participants were told that the vehicle they saw was either manually driven or automated, each vehicle was also manipulated in a manner to enhance the illusion that they were in one of two driving modes (see [Fig. 1](#)):

- Driving mode D1: Manual driving – the car is operated manually by a clearly visible human driver.
- Driving mode D2: Automated driving – the car is driving autonomously in the absence of a clearly visible human operator.

The automated driving mode was created through equipping the vehicles with a Wizard of Oz prototype, applying the Ghost Driver approach ([Rothenbücher et al., 2016](#)). In this approach, the driver wears a “seat suit” while driving the vehicle, giving the appearance from the outside that the vehicle has no driver in it. Apart from this, there were no other markings or stickers on the vehicle to call it out as an experimental vehicle. Furthermore, there were no additional sensors on the vehicle that would call attention to it as an automated vehicle, other than the fact that it appeared that the car was driving itself without anyone being present inside the vehicle. To further distinguish the manual driving mode from the automated



Fig. 1. The Ghost Driver prototype - a “seat suit” was developed for the driver to give the illusion that the vehicle is driving in an automated manner. The vehicle, with and without the seat costume, emulating an automated and manually-driven vehicle respectively, are shown.

one, in the former scenario, the driver wore a high visibility jacket. This made it more easily apparent from outside the vehicle that the vehicle had a human driver. In the other case, the black seat suit, combined with the black interior of the vehicles made the vehicle appear convincingly from outside that it was driving in an automated manner (Fig. 1).

Once the vehicles were equipped with the testing platform, each test vehicle was driven on a straight, otherwise empty road in the university campus. Videos of the approaching car were captured from the perspective of a pedestrian situated on the side of the road (but not at a pedestrian crossing), who has a full view of the road and the vehicle. The camera was mounted on a tripod at eye-level for an average pedestrian (eye height approximately 160 cm). The capturing of the video was started with the vehicles being 100 m away from the pedestrian.

The rationale behind choosing a point on a road in the university campus without a pedestrian crossing was that such crossing behaviour in presence of vehicles is extremely common in this context. Additionally, in The Netherlands – the country where this experiment was conducted, crossing a road in the absence of a pedestrian crossing is not prohibited, nor is the vehicle obliged to give right of way. Pedestrians are simply advised to use caution and discretion while crossing. This setup thus allows for testing the perception of the intention of the approaching test vehicle in this open and flexible environment and context. The pedestrian-perspective view used in the video is shown in Fig. 2.

Eight video footages (2 behaviours from 2 cars with 2 driving modes each) were recorded, as shown in Table 2 below.

The willingness of the pedestrians to cross the road in front of the approaching vehicle was measured at 5 specific points of distances from the pedestrian, the distances being 45 m, 30 m, 15 m, 5 m, and 1.5 m from the pedestrian. Each of the 8 videos of the approaching car was clipped at specific timestamps corresponding to when the vehicle was at these 5 measuring points, which yielded 40 small video segments (8×5) as the stimuli for the experiment. Each video segment stimulus thus showed the approaching car with the video starting by showing the vehicle approaching from 100 m away, and until the point the video was clipped with the vehicle at the measuring point. The video segments were between 4 and 13 s long, allowing participants to form an opinion of the external-appearance-related characteristics and driving behaviour of the vehicle. The relative positions of the pedestrian (camera placement) and the corresponding distances of the 5 measuring points are shown in Fig. 3.

2.2.3. Participants

The experiment was conducted as a between-subjects study between the two driving modes, with 30 participants per condition (M1 through M30 for manually-driven mode, and A1 through A30 for the automated mode). To minimize variability, the target demographic was fixed on people who had normal agility/mobility (as this might affect their decision to cross the road at a given time). Sixty participants between 20 and 55 years in age were recruited for the experiment (mean = 27.6, standard deviation = 6.53). Each participant had either normal or corrected-to-normal vision. Care was taken that the participants recruited were among the students and staff of the university (from an internal participant database) and were familiar with the setting. Since the setting chosen was one without a pedestrian crossing where an approaching car does not necessarily have to yield to a pedestrian, it was important to ensure that the participants had the same knowledge of the area and familiarity with the customary practice of accepting an invitation to cross the road when an approaching vehicle yields the right-of-way.

Only people who did not have experience with automated driving systems were chosen to participate in the automated driving condition. The participants were asked beforehand whether they were involved in any way with work or research in the domain of automated vehicles, or if they had ever interacted with a car with self-driving or advanced automation capabilities. Only if they answered “no” to these questions, they were considered eligible to participate in the condition with the automated driving mode. Otherwise, they participated in the manual mode. This was done to get an even assessment of people's interaction with automated vehicle specifically from people who had no background or previous experience with such vehicles.

2.2.4. Procedure

In each experiment condition, the participant was informed in advance that they will be providing their feedback regarding crossing an unmarked road in the presence of an approaching vehicle. Additionally, the participants were not informed

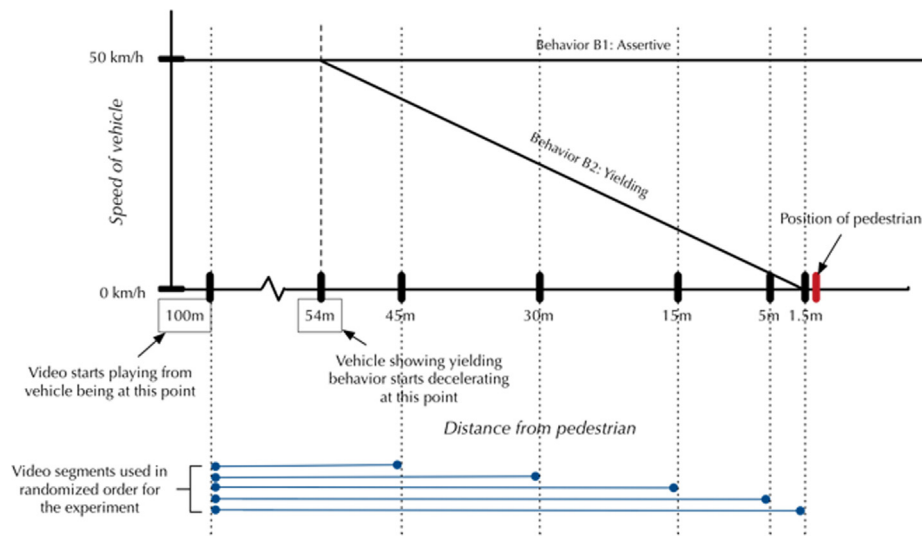


Fig. 2. Screenshots of video stimuli of approaching cars from a pedestrian's perspective as used in the experiment.

Table 2

The combination of video footages used to compare the vehicles, behaviours, and driving modes. In Manual Driving, the driver wearing a high-visibility jacket was clearly visible inside the car. In (unsupervised) Automated Driving, the Ghost driver platform was used, and participants were explicitly instructed that they will encounter an AV.

Vehicle	Behaviour	Driving Mode
V1: BMW 3 series Commonplace and ordinary-looking	B1: Non-yielding	D1: Manual driving D2: Automated driving
	B2: Yielding	D1: Manual driving D2: Automated driving
V2: Renault Twizy Futuristic and alternative-looking	B1: Non-yielding	D1: Manual driving D2: Automated driving
	B2: Yielding	D1: Manual driving D2: Automated driving

**Fig. 3.** Relative positions of pedestrian and measuring points.

whether or not the vehicles were equipped with emergency-braking system in order to deter reckless behaviour. After signing an informed consent form, the participant was asked to sit in front of a 29" widescreen monitor as the video of the approaching car from the pedestrian point-of-view was played full-screen to them. The participants were instructed to imagine that they were standing at the edge of an unmarked road with the intention to cross the road. In the experiment condition involving the Driving mode D2 (automated driving), the participants were explicitly informed in advance that the cars they would be encountering in the video are automated vehicles: that the vehicle drove itself, and that there was no human driver controlling it. No participant who participated in the automated driving condition was given any additional information about the safety of the vehicle. Before starting the actual experiment, the participant was asked to practice with 5 random stimuli to familiarize themselves with the setup. Subsequently, each video clip corresponding to the 5 measuring points for one particular driving condition (manual or automated driving), showing 2 cars of different appearances (BMW 3 series or Renault Twizy), exhibiting 2 different behaviours (assertive or yielding) were presented 2 times in a randomized order to the participant. Thus, each participant was presented with 40 stimuli ($5 \times 2 \times 2 \times 2$). The order of the presentation of the stimuli was randomized to counterbalance any learning effects. After each video segment completed playing, the participants were requested to express their willingness to cross the road at the moment the video had stopped playing on a 5-point Likert scale, with 1 being no willingness and 5 being total willingness by pressing the corresponding key (1–5) on a keyboard.

The willingness to cross the road was chosen as the single measure in this study as is a suitable substitute measure of the feeling of safety or trust of a pedestrian in crossing the road (Walker, Dey, Martens, Pfleging, Eggen, & Terken, 2019). A pedestrian will likely not cross a road if they do not feel safe and have enough confidence that a vehicle will stop for them.

After the experiment, the participants were asked to answer a few questions regarding themselves, and their experience during the experiment. A validated assertiveness questionnaire was applied to gauge the participants' personality and their overall assertiveness, with the conjecture that an individual who is assertive in general might also demonstrate an assertive attitude while crossing a road as a pedestrian. For this purpose, the 30-item schedule for assessing assertive behaviour

(Rathus, 1973) was adapted and augmented with a few related questions regarding behaviour in traffic situations to contextualize the responses. Additionally, a small semi-structured interview was conducted where the participants were asked questions about their general opinion about automated vehicles. In the experiment condition involving manually-driven vehicles, they were asked to speculate if and how they would expect their responses to change if the vehicle in question was automated. In the experiment condition involving automated vehicles, they were asked to reflect how the knowledge that the vehicle is automated affected their decision to cross the road. The entire process including the instruction, the experiment, and the post-experiment interview ran for a total duration of 30–35 min. Each participant was compensated for their time with a €5 gift card.

3. Results

3.1. Quantitative

The experiment dealt with 3 different independent variables across two groups of participants in a mixed design: vehicle appearance (BMW 3 Series, Renault Twizy), vehicle behaviour (non-yielding, yielding) as within-subjects manipulations, and vehicle driving mode (manual, automated) as a between-subjects manipulation. The dependent variable was the willingness of a pedestrian to cross the road in the presence of the oncoming vehicle. An inspection of descriptive statistics showed that the behaviour of an approaching vehicle had a profound effect in pedestrian willingness to cross in front of it. The mean scores of pedestrian willingness to cross in Table 3, and the graphs in Fig. 4 show the impact of the three different aspects of the vehicle (appearance, behaviour, and driving condition) on the pedestrian willingness to cross, at different distances.

It is observed that the general pattern of pedestrians' willingness to cross does not change substantially between the two different vehicles (BMW 3 series vs. Renault Twizy), or between the two different driving modes (manual vs. automated), but rather conforms based on the behaviour of the vehicles. In the case of a vehicle exhibiting non-yielding driving behaviour, the willingness to cross the road diminishes steadily as the vehicle comes closer, which is an expected response. When a vehicle exhibits yielding behaviour, the willingness of the pedestrians to cross diminishes as the vehicle comes closer until a certain point. When the vehicle is close enough that the pedestrian can be relatively sure that the car is stopping, or when the car has come to a complete stop, the willingness to cross increases again.

Analyses of variances (Mixed ANOVA) were conducted for each measuring point (45 m, 30 m, 15 m, 5 m, and 1.5 m), with behaviour and vehicle appearance as within-subjects factors, and driving condition as a between-subjects factor, and the pedestrians' willingness to cross as the dependent variable, to evaluate the main effects and the two-way interactions and the three-way interaction between the independent variables. In each case (distance of vehicle from the pedestrian), the assumption of sphericity (verified by Mauchly's Sphericity Test significance value >0.05) holds when looking at the F-statistics. For the sake of brevity, only the main effects are reported in Tables 4 and 5, as the interactions were not significant.

Results show a highly statistically significant main effect of vehicle behaviour on a pedestrian's willingness to cross at every measuring point (Table 4). We see a clear difference in the willingness of the pedestrians to cross in front of the vehicle as it approaches closer between the assertive and yielding behaviours. Pedestrians consistently showed a higher willingness to cross in front of an approaching vehicle which demonstrated a yielding behaviour, when compared to one demonstrating an assertive behaviour, which is intuitive and expected behaviour.

Table 3

Pedestrian willingness to cross across variations of vehicle driving condition, appearance, and behaviour. 1 = Totally unwilling to cross; 3 = undecided (neutral); 5 = Totally willing to cross Top: Mean score; Bottom (small italics): Standard deviation.

Vehicle	Behaviour	Driving Mode	Vehicle distance from pedestrian				
			45 m	30 m	15 m	5 m	1.5 m
BMW 3 series	Non-yielding	Manual	3.683	2.933	1.817	1.450	1.25
			<i>1.141</i>	<i>0.907</i>	<i>0.782</i>	<i>0.723</i>	<i>0.583</i>
		Automated	3.2	2.35	1.533	1.2	1.083
			<i>0.702</i>	<i>0.789</i>	<i>0.434</i>	<i>0.484</i>	<i>0.296</i>
	Yielding	Manual	3.733	3.617	3.467	3.617	4.567
			<i>0.935</i>	<i>0.962</i>	<i>0.964</i>	<i>1.142</i>	<i>0.907</i>
		Automated	3.667	3.633	3.75	3.733	4.667
			<i>0.735</i>	<i>0.798</i>	<i>0.998</i>	<i>0.989</i>	<i>0.592</i>
Renault Twizy	Non-yielding	Manual	3.433	2.567	1.633	1.5	1.317
			<i>1.158</i>	<i>1.023</i>	<i>0.765</i>	<i>0.841</i>	<i>0.676</i>
		Automated	3.15	2.4	1.517	1.183	1.05
			<i>0.939</i>	<i>0.855</i>	<i>0.663</i>	<i>0.482</i>	<i>0.153</i>
	Yielding	Manual	3.85	3.45	3.167	3.383	4.533
			<i>0.984</i>	<i>0.864</i>	<i>0.874</i>	<i>0.971</i>	<i>0.850</i>
		Automated	3.683	3.1	3.2	3.717	4.733
			<i>0.759</i>	<i>0.904</i>	<i>1.087</i>	<i>1.023</i>	<i>0.521</i>

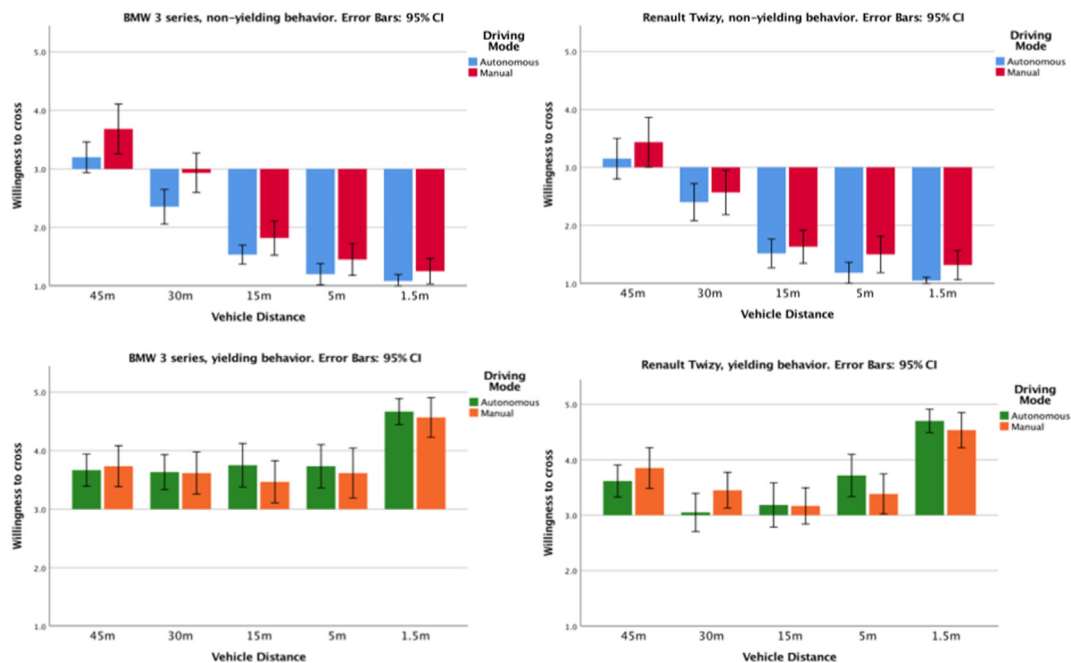


Fig. 4. Mean scores of pedestrian willingness to cross the street (1 = Totally unwilling to cross; 3 = undecided (neutral); 5 = Totally willing to cross) in the presence of the oncoming vehicles exhibiting different behaviours and driving modes at various distances from the pedestrian.

Table 4

Main effects of the driving mode and behaviour of a vehicle on pedestrian willingness to cross.

Vehicle Distance	F (1, 58)	Significance	Effect Size (η^2)
<i>Behaviour: Yielding vs. Non-yielding</i>			
45 m	20.349	<0.000	0.260
30 m	92.832	<0.000	0.615
15 m	282.638	<0.000	0.830
5 m	349.248	<0.000	0.858
1.5 m	1306.303	<0.000	0.957
<i>Informed Driving Condition: Manual vs. Automated</i>			
45 m	1.462	0.232	0.025
30 m	2.385	0.128	0.039
15 m	0.017	0.896	<0.000
5 m	0.030	0.862	0.001
1.5 m	0.099	0.754	0.002

Inspection of the descriptive statistics in Table 3 and the graphs from Fig. 4 does not show a consistent pattern of influence of pedestrians' road-crossing willingness based on the driving mode of the vehicle. Additionally, the mixed ANOVA results (Table 4) failed to show a statistically significant difference in pedestrian responses between the manual and automated driving conditions at any measuring point. The tests also show that the driving mode gained a very small effect size, which further corroborates the inference that knowledge of the driving mode of the vehicle did not affect the way pedestrians interacted with the vehicle.

Table 5

Main effects of the vehicle appearance on pedestrian willingness to cross.

Vehicle Appearance: Renault Twizy vs. BMW 3 series			
Vehicle Distance	F (1, 58)	Significance	Effect Size (η^2)
45 m	0.351	0.556	0.006
30 m	7.419	0.009	0.113
15 m	4.732	<0.000	0.203
5 m	0.830	0.366	0.014
1.5 m	0.116	0.735	0.002

Table 6

Spearman's correlation coefficient between participant assertiveness and their willingness to cross.

	Manual	Automated
45 m	$\rho = 0.171, p = 0.031$	$\rho = 0.096, p = 0.147$
30 m	$\rho = 0.010, p = 0.457$	$\rho = 0.109, p = 0.118$
15 m	$\rho = 0.055, p = 0.274$	$\rho = 0.009, p = 0.462$
5 m	$\rho = -0.103, p = 0.132$	$\rho = -0.005, p = 0.478$
1.5 m	$\rho = -0.069, p = 0.228$	$\rho = -0.017, p = 0.425$

The effects of the different vehicles are statistically significant only within a certain range of distance from the pedestrian (Table 5). This is seen as a marked difference in the participants' responses at the measuring point of 30 and 15 m from the pedestrian. This difference may be attributed to the contrasting appearances of the vehicles. In this range of distance (measuring points of 30 m and 15 m), pedestrians appeared more willing to cross in front of the BMW 3 series compared to the Renault Twizy. This was observed nearly consistently in both manual and automated driving modes.

It was also investigated whether the individual assertiveness of the participants had any role to play in their willingness to cross the street, in order to account for the role of assertiveness as a potential covariate. The assertiveness scores of each participant (Rathus, 1973), were assessed against their responses, and the results of the Spearman's ρ is shown in Table 6. Analysis shows that in each measuring point, the assertiveness of the pedestrian had either negligible or weak correlation, and in all but one scenario (manual driving, measuring distance of 45 m), the correlations were not statistically significant. As a result, it can be inferred that for practical considerations, individual assertiveness did not play an active role in determining the willingness of the pedestrian to cross, and the attributes of the vehicle were more significant determinants of pedestrian behaviour.

3.2. Qualitative

While quantitative data shows how people acted, it does not explain the subjective reasons behind those actions. Additionally, the way people act is often different from how people think they act, so it is still important to capture an account of conscious rationale behind the perceptual and behavioural characteristics of a user. To this end, the qualitative data from the post-experiment semi-structured interviews with the participants were subjected to thematic analysis and categorized. We looked for various elements in the participant responses that alluded to factors that influenced their decisions to cross the road in the presence of the oncoming vehicles. The themes that emerged from the analysis are discussed below and are additionally furnished with a few selected quotes by the participants which exemplify the insights.

3.2.1. Behaviour of the vehicle is the most critical determinant of willingness to cross

When asked the open-ended question of what factors influenced their willingness to cross the road in front of the approaching vehicle in the experiment, every participant stated that the distance of the car played the most crucial role while deciding to cross the road. This is an expected response: if the car is still far away, pedestrians will simply cross the road. However, the situation becomes increasingly ambiguous as the car comes closer. The participant responses were also unanimous across both driving modes (manual vs. automated) that apart from the distance, the vehicle speed and behaviour played a primary role in determining their willingness to cross the road. The movement patterns of the car (whether the car seemed to be slowing down, or continuing to drive at speed) impacted their decision-making process, as revealed in the following relevant participant quotes:

[M4]: "I first focused on the speed at which the car is approaching. If I feel that the speed is reducing and it is coming to a stop, I decided to cross the road, otherwise not. The speed, acceleration, and deceleration were important. In the videos, I wasn't able to see the driver very well. But in real-life scenarios, at whatever speed the car is approaching, if I am able to see the driver and establish eye contact, I will cross the road. But that is a secondary confirmation. The primary factor is the speed and the behaviour of the vehicle. If I am confident that the speed is decreasing, I will cross the road."

[M22]: "I was looking at the speed of the car and how far it is from me. If the car is too far away, I would just cross and I don't even need to look at the speed."

[A2]: "If the car stops before me, I will cross, otherwise I will be reluctant. I don't trust the car while it is still in motion. If the car is far away, I look at the speed of the car. Even if the car is far away, if the car is driving quickly, I will just wait."

[A28]: "When the car came across the first time, I looked inside and saw that there was no one driving the car. Since then, I realized that "oh, I don't need to look for the driver if they see me or not", so I was just looking at the speed of the car".

3.2.2. Size matters: the perceived importance of vehicle size in the crossing decision

When the participants were asked if the difference in the cars they encountered had any effect on their crossing decisions, the majority responded that the appearance of the car did play a role in their decisions. Within this group of participants, a majority in both driving modes highlighted the size of the vehicle as being a factor in making a decision. The observable size

difference between the Renault Twizy and the BMW 3 Series, played a conscious role for most people, and they rationalized that with a smaller car, they can cross and get out of the car's way more quickly, so they were more likely to cross in front of a small car compared to a big car. Alternatively, in the worst case if they get hit by the car, a smaller car would cause less damage. There was a clearly higher perceived risk in interacting with larger vehicles. The subjective consideration of vehicle size is evidenced by some of the representative participant quotes below:

[A1]: "I was thinking about the looks of the car. I think that when I saw the bigger car, I inherently responded with a little more caution. I feel like 'I see a bigger car, I need to be more careful'. I feel that the bigger car has more respect (presence) on the road. At the same speed, I would feel more afraid to cross in front of the bigger car."

[A3]: "I felt more scared around the bigger car. I felt that the bigger car was driving faster than the smaller car. Also, if the car is small, I can avoid it by moving quickly."

[M3]: "If the car is large, I think it would take a longer time to slow down, so I feel more threatened by it, even if it is farther away. I also feel that if the smaller car hits me, maybe I'll get a bruise, but if the bigger car hits me, I'll probably be dead."

[M26]: "I know that theoretically, the difference in the vehicles shouldn't matter. But the fact that the BMW was a bigger car made me want to keep a safer distance from it when compared to the other car. I was being more cautious around the BMW."

However, it is interesting to note that this contrasts the actual responses made by the participants, as found in the quantitative analysis. The actual willingness to cross was not affected by the perceived risk. We revisit this in the discussion.

3.2.3. Stereotypes matter: mental models and preconceived ideas about vehicles and their driving styles influence crossing decisions

Stereotypes of vehicles based on their external appearance, and subsequently, their expected behaviours did play a role in the conscious reasoning of some participants. While these preconceived notions of various participants did not necessarily concur with each other, or conform to the truth, they sometimes had a conscious effect on the participants. This, paired with the quantitative results indicated that the appearance of the vehicle and associated stereotypes and prejudices might play a role in pedestrians' crossing decisions.

[M9]: "I feel that the Twizy is an odd vehicle. That's not something you would encounter regularly on the road. I would consider the BMW to be a more 'normal' vehicle. . . I don't necessarily find the Twizy cute or friendly. Moreover, if it's going fast and it hits you, it is still going to be a threat. But if both the vehicles are traveling at the same speed, I'd trust that the BMW would brake better than the Twizy. . . I don't think whether a car looks cute matters. However, it also interferes with the thought that the Twizy being smaller is easier to avoid if I move quickly, while the BMW might still hit me because it is a wider, bigger car. At the end of the day, it depends on how fast the car is coming, and how far they are. I was weighing those options to determine if it was safe for me to cross."

[A14]: ". . . for the different look (speaking about the Renault Twizy), I didn't feel confident every time as I didn't know how to react around it.

[A25]: "The small car (the Renault Twizy) looked more sophisticated, intelligent, and futuristic, so I thought it might stop sooner or be better equipped with handling an emergency braking situation. So, I might be bolder to cross the road."

[A27]: "The BMW made me more reluctant to cross. BMWs are normally driven by people who drive aggressively, and they drive faster. I didn't think differently because this was an automated car. I wanted to be careful."

[A29]: "I actually found the small car a little more unpredictable because it's a car I am not familiar with seeing. It reminds me of a brommobiel (mini car), and the people who drive those things are in my experience more unpredictable than people driving normal cars."

3.2.4. Unique and futuristic looks make a car more believable as an automated vehicle

In the majority of the cases, a futuristic or unique external appearance in a vehicle did lend itself to the idea that the vehicle is equipped with advanced technology and is capable of automated driving, as expected and desired in our experiment design. In the automated driving condition, some participants indicated that it was easier for them to believe that the Renault Twizy was an automated vehicle because it was more futuristic and "robotic-looking" [A15]. Even in the experiment condition involving the manual driving mode, one participant [M1] commented that the uniqueness of the Renault Twizy made them think that the vehicle was driven by a robot (thus self-driving), despite it being driven by the human driver in a high-visibility jacket. On the other hand, the BMW was a familiar vehicle and conformed to people's idea of an ordinary car, so it was less obvious as a self-driving car.

[A8]: "When I saw the smaller car (the Renault Twizy), I really thought that it was a self-driving car. I don't know why. But in my mind, the first impression was that the other (BMW) did not look like a self-driving car."

[A9]: "The small car looked like it was equipped with new technology and innovation. The normal one (the BMW 3 series) – it looked like cars with drivers in it. As it was said that they are both self-driving cars, it didn't influence my decision to cross the road. But if I just look at the car, the small one looked more hi-tech."

[A15]: "The big car didn't really look like a self-driving car. The other car was remarkably smaller than you are used to seeing. It looked a bit more robotic; I think. A bit more mechanical than the big car."

[A25]: "The small car looked more sophisticated, intelligent, and futuristic, so I thought it might stop sooner or be better equipped with handling an emergency braking situation. So, I might be bolder to cross the road. I caught myself thinking that

the BMW was just an ordinary car with someone driving it, and not an automated car. The small car was more believable as an automated car for me.”

[M1]: “The small car (Renault Twizy) looks like one that is used for experiments. I wouldn’t want to drive around in that car because it will be difficult to be seen in. As the car looked a little strange, I didn’t know its behaviour and I couldn’t predict how it was going to slow down or stop. For the bigger car (the BMW 3 series) – I have seen such cars around and I know what to expect. But with the small car, I couldn’t be sure. Somehow, it felt that instead of a person, there was a robot inside the car because in my mind I associated the car with experimental prototypes. That’s also why I wasn’t entirely sure if the car would really stop for me”

3.2.5. Explicit knowledge of the self-driving nature of a vehicle has little impact on crossing behaviour

As a response to whether the knowledge of the vehicle being automated consciously affected their decision, a few participants reflected that the knowledge that a vehicle is automated would either fill them with hesitation, or contrarily, confidence. However, a majority of the participants denied the self-driving nature of the vehicles to have had any impact on their crossing decision.

[A1]: “If it is an automated car, I would not like to step in even if it is slowing down. But in this case, I didn’t think about it too much. I think I tuned out of the fact that these were automated cars”.

[A8]: When I saw the smaller car (the Renault Twizy), I looked inside and saw that there was no driver inside. Then I realized that it was a more hi-tech self-driving car, and it made me feel more confident around it. (This participant also said that the BMW did not look like a self-driving car in their opinion – see quote above).

[A2]: I imagined myself in a world where automated cars have been tested and approved to be on the road, so they must be safe. I didn’t think I acted any differently as opposed to a normal (manually-driven) car.

[A12]: I was just basing my decision on the speed and distance of the cars, so I don’t think that the fact that they were self-driving mattered.

Subjective opinion was split regarding the perceived safety of automated vehicles. However, the actual road-crossing willingness was not affected by individual differences in regards to perception towards automated vehicles, as seen from the quantitative analysis: the unaided knowledge of the driving mode did not cause a significant departure in the pedestrian willingness to cross. This strongly points to the inference that simply introducing a reference frame that a vehicle is automated does not adversely impact pedestrians’ attitude towards such vehicles.

4. Discussion

In this experiment, the effects of a vehicle’s driving mode (automated vs manual), external appearance, and driving behaviour (yielding vs not yielding) on the willingness of pedestrians to cross a road in its path were studied. The primary aim was to identify whether pedestrians behave differently with automated vehicles as compared to manually-driven vehicles.

4.1. Driving mode

No evidence was found that the knowledge whether a car is self-driving or manually-driven influences pedestrian interactions with the vehicle. Evidence from the post-experiment interviews is in alignment with this insight. Based on this, we reject our hypothesis H1 (*The perceived driving mode of the vehicle: manual vs automated, affects the willingness of pedestrians to cross the street*). This finding confirms the outcomes of an earlier study reported in the literature (Rodríguez Palmeiro et al., 2018), and suggests that people are not more uncertain about whether to cross in front of an automated vehicle, or – on the contrary – have more trust that an automated vehicle will halt.

4.2. Vehicle driving behaviour

On the other hand, the results showed that overall, the behaviour and driving dynamics of the vehicle play a significant role in implicitly communicating the intent of the vehicle to pedestrians. When a vehicle exhibited non-yielding driving behaviour, the response was unsurprisingly and consistently a decrease in the pedestrians’ willingness to cross as the vehicle came closer. In contrast, when the vehicle exhibited yielding behaviour, it was observed that the pedestrians’ willingness dropped as the car came closer until a certain point, and then the willingness increased again. This is explained by the fact that when the vehicle is far away, the distance of the vehicle inspires enough confidence for the pedestrian to cross. As the vehicle gradually slows down when approaching the pedestrian, the situation becomes increasingly ambiguous and the pedestrian becomes less confident that it is safe to cross. When the vehicle still gets closer and reduces speed further, the pedestrian’s confidence that the vehicle will stop and that it is safe to cross, begins to increase again. This is in alignment with hypothesis H2 that the driving behaviour of the vehicle plays a significant role in pedestrians’ road-crossing decisions. The qualitative data also indicate that pedestrians rely on the vehicle’s driving dynamics and movement patterns substantially to inform their road-crossing decisions.

In previous work, Zimmermann and Wettach (2017) conducted an exploratory study in which suggested that a vehicle's motion behaviour was able to evoke certain emotions, which in turn lead to rational decisions within the participants. The authors argued that this can be seen as a form of mutual communication. Similar findings are observed in the research by Risto, Emmenegger, Vinkhuyzen, Cefkin, and Hollan (2017), which show that drivers in manually-driven vehicles often choose to exaggerate their behaviour in traffic, such as stopping short, to explicitly show intention through motion patterns. Other work has shown that braking patterns and subsequent time-to-arrival estimates can act as a basis of communication of intent for vehicles towards pedestrians (Ackermann, Beggiato, Bluhm, & Krems, 2018; Ackermann, Beggiato, Bluhm, Löw, & Krems, 2019; Beggiato, Witzlack, & Krems, 2017; Beggiato, Witzlack, Springer, & Krems, 2017). The findings in this study corroborate the earlier insights: the motion patterns and driving behaviours of a vehicle have profoundly significant effects in communicating the vehicle's intention, and emerges to be the biggest determinant of pedestrians' willingness to cross, especially when the vehicle clearly exhibits yielding and non-yielding behaviour. Further research is required to investigate pedestrian interaction with a variety of different vehicle motion patterns to validate whether the movement behaviour of the vehicle is a significant contributor of pedestrian road-crossing decision in every scenario.

4.3. Effect of external appearance

The main effect of appearance showed small but statistically significant differences at measuring points in the intermediate distance range (30 m, and 15 m). Research with conventional (manually-driven) vehicles has shown that the external appearance of the vehicle reflects the temperament and driving style of the driver and lends itself to stereotyping (Davies & Patel, 2005; Davies, 2009). One might assume that since automated vehicles are driven by computer algorithms and not by people, the vehicle appearance will cease to have an effect on pedestrians' willingness to cross a road in its presence. However, evidence from this study suggests otherwise. At the measuring points where the vehicle appearance becomes significant (30 m and 15 m), in both manual and automated driving modes, we notice that the pedestrian willingness to cross is lower in front of the Renault Twizy compared to the BMW 3 series.

One aspect of the vehicles' appearances that may have played a role in the pedestrians' willingness to cross is the size difference of the vehicles. From the qualitative results, it also emerged that pedestrians consciously felt the need to be more cautious around a larger car compared to a smaller car. However, we observe from the quantitative data that in reality, the opposite happens, and that pedestrians are less willing to cross in front of the Twizy compared to the BMW. This might be explained by the size-arrival effect, which holds that smaller vehicles are perceived to be farther away than larger vehicles, so that the estimated time to arrival for smaller vehicles is later than for larger vehicles (DeLucia, 1991, 2013; Levulis, DeLucia, & Jupe, 2015; Smith, Flach, Dittman, & Stanard, 2001).

A second aspect may have been the vehicles' perceived sociability. A priori, pedestrians were expected to have a lower willingness to cross in front of the BMW 3 series, compared to the Renault Twizy, at the same distance and driving condition. The underlying reasoning was that the BMW scored as a more aggressive, angry-looking vehicle than the comparatively cute and friendly-looking Twizy. Existing research suggests that the exterior appearance of a vehicle, specifically, the shape of its headlights and grills, plays a significant role in the perception of the vehicle's sociability and power, or friendliness vs. aggressiveness (Landwehr, McGill, & Herrmann, 2011; Windhager et al., 2008, 2012). This, in turn, was expected to affect pedestrians' crossing behaviours in front of cars (Klatt et al., 2016) by lowering the willingness to cross in front of aggressive-looking vehicles. However, as mentioned above, the results showed that pedestrians were more willing to cross in front of the approaching BMW than the Renault Twizy.

Combined, these lead us to reject hypothesis H3a – the smaller size or friendlier appearance of a vehicle did not inspire a higher willingness for pedestrians to cross. Furthermore, we also observe that pedestrians' responses were similar across both manual and automated driving modes. Thus, we reject hypothesis H3b – the vehicle appearance still played a role for automated vehicles.

This brings us to the third dimension of the appearance of the vehicles: their novelty and uniqueness. The BMW was perceived as a commonplace, ordinary-looking vehicle, as opposed to the Twizy, which was perceived as an alternative, futuristic-looking vehicle. While the unconventional appearance (uniqueness, conspicuity, and futuristic-ness) of the Twizy settled into the belief for many participants that the car is equipped with advanced technology and capable of automated driving (as evidenced by the participant interviews discussed in the qualitative analysis), the results suggest that subconsciously, the peculiarity and the unfamiliarity of the appearance of the Renault Twizy may have adversely affected the participants' trust in its technical capabilities. People felt more confident in determining the behaviour of the more familiar, albeit aggressive-looking BMW that was more believable as an ordinary, manually-driven vehicle.

4.4. Vehicle appearance as an indicator of self-driving capability

An interesting aspect of the general propensity of people to consider the BMW 3 series as an ordinary car, and contrarily, the Renault Twizy as a more believable automated vehicle, is the fact that this behaviour transpired despite the participants having explicitly been informed of the vehicles' automated nature in the experiment instructions. We infer that when a car is commonplace in looks, and does not have markers or external artefacts that explicitly draw attention to its automated driving capabilities (e.g. stickers proclaiming that it is an automated vehicle, or sensors clearly visible on the vehicle that mark the car out as a departure from the norm), people may lose a conscious cognizance of the fact that the car is self-driving, and

regard it as any other normal, human-driven vehicle. The question arises whether the outcome would have been different if the “ordinary” vehicle in the experiment (BMW) was marked or equipped with sensors in a manner that called attention to its self-driving nature.

In a prior study on the perceptions of an automated shuttle to be used for public transport, [Schieben, Wilbrink, Madigan, Louw, and Merat \(2018\)](#) conducted face-to-face interviews, an onsite-survey and two focus groups resulting in several design considerations. In the context of the perception of a vehicle as having automated driving capabilities, their participants pointed out that it is important for the design of the vehicle to look unlike normal public transport systems to be clearly identified as an automated vehicle. Their findings suggested that being able to identify the shuttle as an automated vehicle helps to build up the right expectations on the behaviour and next manoeuvres of the vehicle. However, taking into account the evidence from our study, we question whether vastly different exterior designs in vehicles is a good in all circumstances. Rather than helping pedestrians to build the right expectations, a futuristic exterior may make them actually less confident about how the vehicle will behave.

4.5. Implications

The biggest takeaway of this study is that all other factors remaining constant, pedestrians do not experience a significant change in their willingness to cross in front of an automated vehicle as opposed to a manual vehicle. The external appearance of the vehicle has a small modulating effect on willingness to cross. A futuristic appearance seemingly reinforces the instruction to the participants that a vehicle is automated.

Our results show an impact of vehicles' external appearances on pedestrians' perception of a vehicle as automated, and their willingness to cross a road. The patterns that emerge out of this data – particularly regarding the novelty and futuristness of a vehicle's appearance – lay the foundations of further research investigating the effect of vehicle appearance in more scenarios for automated vehicle deployment. These lead to a major design insight. Given that car manufacturers are already choosing to follow different schools of thought in the matter of how the design of an automated vehicle can play a role in conveying the subtle messages behind a car's personality, capability, and power, this research informs the community of future design decisions. Volvo has chosen to make their first test automated vehicles look inconspicuous and like any other ordinary vehicle, to address the concern of invoking abusive or bullying behaviour from other road users – the knowledge that a vehicle is automated by its special appearance, coupled with the assumption that it is programmed for defensive behaviour might embolden pedestrians to take advantage of it by stepping in its front even in the absence of the right of way without any fear of consequences ([Connor, 2016](#)). In contrast, the Google self-driving car was purposefully designed to look cute and friendly in order to help pedestrians regard it as a harmless entity, as well as impart the sense that it is not a high-speed vehicle ([D'Onfro, 2014](#); [Korosec, 2016](#)). Other car manufacturers employ special design techniques to help their concept vehicles stand out. Faraday Future, for instance, calls out its vehicle FF91's design and draws attention to its futuristic features ([Faraday Future, 2016](#)). This research suggests that an overly odd and unfamiliar exterior design of vehicles might interfere with the cognitive models of pedestrians and make them more hesitant in a road-crossing context, at least until the novelty wears off. *Especially in the early stages of introduction of automated vehicles, in order to ensure a smooth introduction and integration into society, it might be important to not make automated vehicles too conspicuous by design.*

Furthermore, we found evidence that the behaviour and movement patterns of a car play the most important role in determining pedestrian crossing behaviour. The impact of their outer appearance is not trivial, but less substantial in comparison. This points towards the design implication that ways to augment the implicit communication that takes place through the movement patterns of a vehicle should be explored for the most effective method to communicate vehicle intent. The most intuitive way to express the intention of a vehicle is through behaviour and movement. Augmenting the vehicle's natural behaviour may be a more effective way to demonstrate vehicle intent than augmenting vehicle appearance using lights, messages, and projections. Communication via deliberate exaggeration of movement patterns is perhaps a better communicator of intent, and more research needs to be conducted on how this can be accomplished.

4.6. Limitations and future work

One consideration of the study is that we manipulated the belief of our participants regarding the self-driving nature of the car in the autonomous driving condition by instruction in the automated driving condition. We did not do an explicit manipulation check to validate whether the belief was affected. In the semi-structured interview after the study, participants were asked how their road-crossing willingness was affected as a result of the vehicles' automated nature. Although no participants mentioned doubting the automation status of the vehicle, we cannot be certain how well the manipulation worked for each participant.

Another limitation of this study was the video-based nature of the experiment as opposed to a real-world field investigation. Due to the lack of the immediate danger of physical harm, participants might have given responses that would have entailed riskier behaviour than they would actually display in real life scenarios. Also, the video-based nature of the experiment limits the dynamic situation of road-crossing behaviour in real life, where the driver/vehicle also reacts to the behaviour of the pedestrian. However, the requirement of a high level of control in the environment and the independent variables made it realistically impossible to conduct this study in the field, as ensuring consistency in the driving behaviour, environmental conditions, and other dynamic attributes are extremely difficult in this context. Therefore, a deliberate judg-

ment was made to use videos as a medium for this experiment as a calculated trade-off between realism and control. Another concern was whether the approach of the vehicle and perception of the arrival time of a vehicle in real-life holds in a video. However, a previous study found that time arrival estimates do not differ between video and real-life situation (Recarte et al., 2005), which retains the ecological validity of this study. Furthermore, in road-crossing situations, pedestrians usually walk up to the edge of the road, make an immediate judgement based on the environment and the approaching car, and decide whether or not to cross the road. Looking at an oncoming vehicle for a prolonged period of time and then indicating road-crossing willingness is not natural behaviour in this context. However, it was important to study pedestrians' interactions with oncoming vehicles over the entire approach of the vehicle as a function of the vehicle's distance. Thus, this limitation was acknowledged within the study design as a trade-off for a longitudinal evaluation of pedestrians' road-crossing willingness.

Prior research points out that pedestrians' road-crossing behaviour depends on several factors such as the structure of the street, the driver's reaction to the communication or time to collision (Rasouli, Kotseruba, & Tsotsos, 2017). Subsequent work also proposed that there is a strong interrelationship between factors that influence pedestrian behaviour (Rasouli & Tsotsos, 2018). This means that only studying a small subset of these factors may not capture the true underlying reasons behind pedestrian crossing decisions. The study was conducted in a controlled setting, on a very specific scenario that involved the interaction of pedestrians with one car at a time, in a simple traffic situation (straight road; daytime conditions; no other traffic). While the scope for the current study was chosen deliberately to study people's reactions to automated vehicles in regular, predictable, and non-emergency situations, the results might differ in situations involving more complex intersections, mixed traffic, and unpredictable behaviours. Future work must involve investigations in more complex conditions to validate the impact of these effects.

5. Conclusion

Within the scope of the simple and predictable traffic situation in which this study was modelled, the findings in this study indicate the patterns of pedestrian behaviour in relation to a vehicle's perceived driving mode, external appearance, and driving behaviour. This information can give direction to the development of automated vehicles. The results suggest a pattern in pedestrian behaviour in relation to vehicle driving behaviours and external appearance in normal, non-critical and non-emergency conditions. The overarching insight is that in interactions in relatively simple traffic conditions, the knowledge of the driving mode of a vehicle does not appear to impact pedestrian interaction with vehicles significantly: pedestrians are not overly hesitant of automated vehicles. The behavioural dynamics and movement patterns of a vehicle play a critical role in implicitly conveying the intentions of a vehicle to pedestrians. This can be thought of as the "body language" of a vehicle, which is the primary cue of vehicle intent in traffic. This insight leads to a practical recommendation of focusing design efforts of automated vehicle human factors in the direction of using the vehicles' behavioural dynamics as the foundation of a communication platform. Furthermore, we found that at a range of distance from a pedestrian when a car is neither too far away nor close enough, the appearance of the car can play a small but significant role in decision-making. The results suggest that the novelty aspect of a car's design plays a role in affecting the mental model of pedestrians and leads them to draw conclusions regarding whether a car is automated, and this, in turn, has the potential to affect their decision to cross a road in its presence at a certain range of distance. On the other hand, we found evidence that when the external appearances of automated vehicles correspond to people's ideas or expectations of any ordinary, manually-driven vehicle, people may forget that they are interacting with automated vehicles even if they are explicitly primed with this knowledge in advance. This insight may be valuable to consider in the design of automated vehicles in the future in the nascent stages of the introduction of self-driving cars when interactions between automated vehicles and vulnerable road users may be impeded by mistrust of new technology.

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